Reg. No. | | | | | | | | | | |

[14EC662]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE-2019

DATA STRUCTURES USING C++

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least TWO questions from each Part
2. Missing data may be suitably assumed.

PART-A

1. a. Construct a C program to print the given two rational numbers are equal or not equal. Use Euclid's algorithm to reduce the rational numbers into its lowest forms. **08**

b. Compare static and dynamic memory allocations and also define the syntax for different types of dynamic allocation and free of memory. **06**

2. a. Construct an algorithm to evaluate an expression in postfix using stack and evaluate the following postfix expression by plotting stack table: **08**

6 2 3 + - 3 8 2 / + * 2 $ 3 +

b. Convert the following infix expression into Prefix and Postfix expressions. **06**

i. A $ B * C – D + E / F / (G + H)

ii. ((A + B) * C – (D – E)) $ (F + G)

iii. A – B / (C * D $ E)

3. a. Define the rules for the Towers of Hanoi problem. How many total steps are required to solve Hanoi problem with three disks? Illustrate each step with neat sketch. **08**

b. Construct a C program to determine the factorial of a given number using recursive method. **06**

4. a. Assume that the Queue is implemented using Array. Construct the C code snippet for the following: **08**

i. Queue structure, Empty Queue, Initial conditions

ii. Remove an element

iii. Insert an element

b. Given a linear linked list, Illustrate the following with neat sketch: **06**

i. Adding an element to the front of a List

ii. Removing a node from the front of a List.

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PART-B

5. a. Assume that the Linear Linked List is implemented using dynamic variables; Construct a C code snippet for the following operations. **08**

   i. Insert_after (P, x) and

   ii. Delete_after(P , Px)

   Where P is pointer to a node, x is integer variable and Px is pointer to integer variable.

   b. What are the significances of using a header node in a linear linked list and also write the node structure for the following type. **06**

   i. Non-integer Node.

   ii. Non-homogeneous Node.

6. a. Write an algorithm / conditions to determine if a binary tree is strictly binary, complete, and almost complete. **06**

   b. Define Binary Tree? Write the Preorder, Inorder and Postorder sequence for the Binary Tree shown in figure 6.a **04**

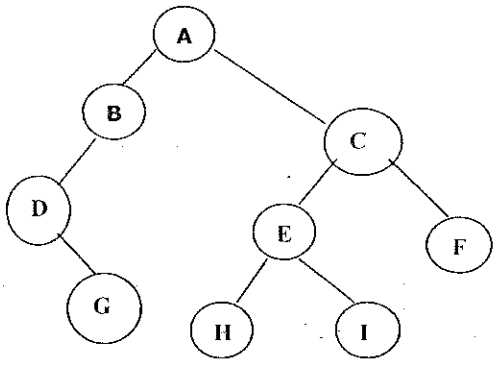

Figure 6.a

   c. Define Binary search tree? Construct a binary tree for the following input list. **04**

   14, 15, 4, 9, 7, 18, 3, 5, 16, 4, 20, 17, 9, 14, 5

7. a. Construct the binary tree for the given unsorted list below. Demonstrate Which type of ordering is used to sort the list in ascending and descending order. **08**

   Unsorted list: **25, 57, 48, 25, 37, 12, 92, 86, 33, 12.**

   b. Given the unsorted list: **25, 57, 48, 37, 12, 92, 86, 33.** **06**

   Demonstrate the algorithms used in sorting the above list in ascending order using bubble sort.

8. a. Illustrate the following. **10**

   i. Inserting into a Binary Search Tree.

   ii. Deleting from a Binary Search Tree.

   b. List the differences between binary and sequential search. **04**